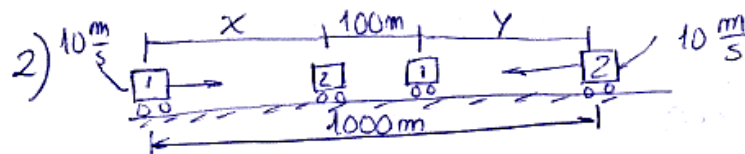


F1



Para 1

$$d_1 = v_1 \cdot t$$

$$x + 100 = 10 \cdot t$$

$$x = 10t - 100 \quad \text{--- } \varepsilon c1$$

Para 2

$$d_2 = v_2 \cdot t$$

$$y + 100 = 10t$$

$$y = 10t - 100 \quad \text{--- } \varepsilon c2$$

Del Dibujo

$$x + 100 + y = 1000$$

$$10t + 100 + 10t + 100 = 1000$$

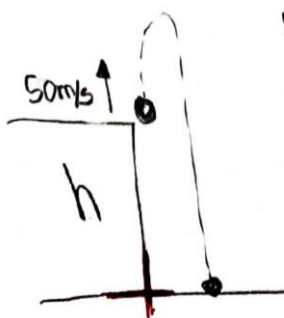
$$20t = 1100$$

$$t = \frac{1100}{20} = 55 \text{ s}$$

$$t = 55 \text{ s} \cdot \frac{1 \text{ min}}{60 \text{ s}}$$

$$t = \frac{11}{12} \text{ min}$$

F2



$$y_F = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$0 = h + 50 \cdot 15 - 5 \cdot 15^2$$

$$h = 375 \text{ [m]}$$

$$y_F = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$0 = 375 - 50t - 5t^2$$

$$t_1 = 5 \text{ s} \checkmark$$

$$t_2 = -15 \text{ s}$$

F3

Diagram of a pulley system with two masses, m_1 and m_2 , and an ideal pulley.

Diagram 1: Shows the pulley system. Mass m_2 is on the left, and mass m_1 is on the right. The pulley is fixed to the ceiling. The tension in the rope is T_1 and T_2 . The acceleration of m_2 is a_2 (down) and the acceleration of m_1 is a_1 (up).

Diagram 2: Shows the ideal pulley. The tension in the rope is T_1 and T_2 . The acceleration of the pulley is a_p .

Equations for the pulley:

$$\sum F_y = m_p a_p$$

$$2T_1 - T_2 = 0$$

$$T_1 = \frac{T_2}{2}$$

Equations for mass m_2 :

$$\Delta y_2 = 2 \Delta y_1$$

$$V_2 = 2 V_1$$

$$a_2 = 2 a_1$$

$$\sum F_y = m_2 a_2$$

$$m_2 g - T_1 = m_2 a_2$$

$$T_1 = m_2 g - m_2 a_2 \quad \text{I}$$

Equations for mass m_1 :

$$\sum F_y = m_1 a_1$$

$$T_2 - m_1 g = m_1 a_1$$

$$2T_1 - m_1 g = m_1 a_1 \quad \text{II}$$

Combining equations I and II:

$$2(m_2 g - m_2 a_2) - m_1 g = m_1 a_1$$

$$2m_2 g - 4m_2 a_1 - m_1 g = m_1 a_1$$

$$a_1 (m_1 + 4m_2) = (2m_2 - m_1)g$$

$$a_1 = \frac{2m_2 - m_1}{m_1 + 4m_2} \cdot g$$

Como $a_2 = 2a_1$:

$$a_2 = \frac{4m_2 - 2m_1}{m_1 + 4m_2} \cdot g$$

con $m_1 = 8 \text{ kg}$, $m_2 = 2 \text{ kg}$ y $g = 9.8 \text{ m/s}^2$

$$\therefore a_1 = -2.45 \text{ m/s}^2 \Rightarrow |a_1| = 2.45 \text{ m/s}^2$$

$$a_2 = -4.90 \text{ m/s}^2 \Rightarrow |a_2| = 4.90 \text{ m/s}^2$$

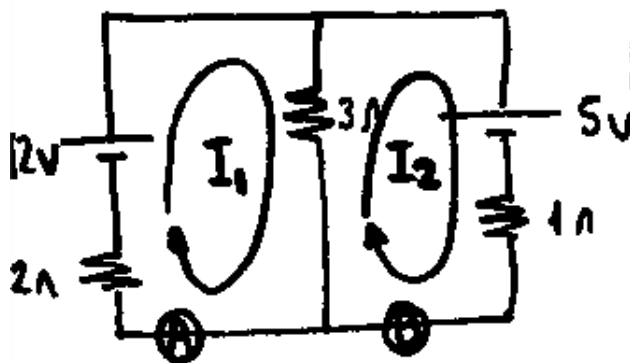
F4

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{8}{R} \Rightarrow R_{eq} = \frac{8}{8}$$

$$R_{eq} = \frac{8}{R} \Rightarrow R = 8R_{eq} = 8 * 2 [\Omega] = 16[\Omega] \Rightarrow R = 16 [\Omega]$$

F5

SOLUCION:



Malla I₁

$$\sum V_f = \sum V_R$$

$$12 = 2I_1 + 3(I_1 - I_2)$$

$$12 = 2I_1 + 3I_1 - 3I_2$$

$$12 = 5I_1 - 3I_2, - \text{EC1}$$

Malla I₂

$$\sum V_f = \sum V_R$$

$$-5 = I_2 + 3(I_2 - I_1)$$

$$-5 = I_2 + 3I_2 - 3I_1$$

$$-5 = -3I_1 + 4I_2, - \text{EC2}$$

solución de EC1/EC2

$$\begin{cases} 12 = 5I_1 - 3I_2 \\ -5 = -3I_1 + 4I_2 \end{cases}$$

$$\begin{cases} 12 = 5I_1 - 3I_2 \\ -5 = -3I_1 + 4I_2 \end{cases}$$

$$\boxed{I_1 = 3A}$$

$$I_2 = 1(A)$$